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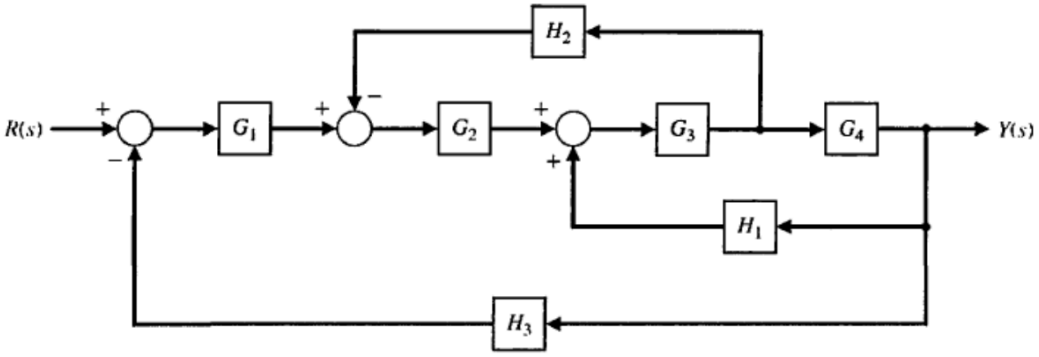
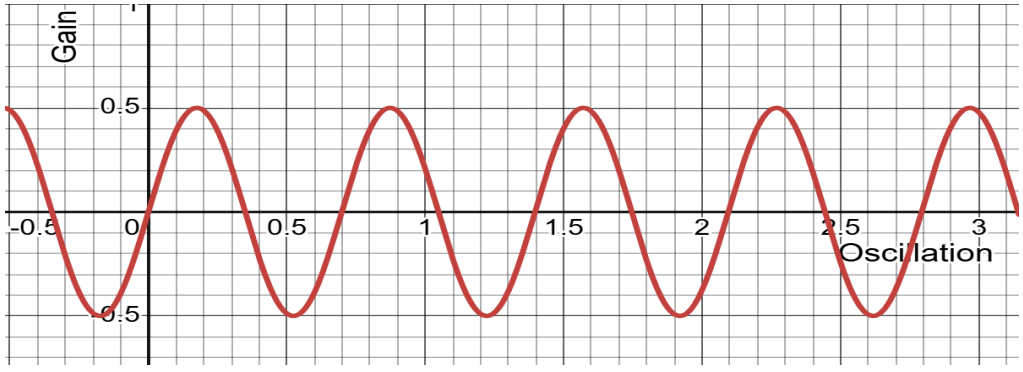
Department of Computer Science and Engineering

CSE 461: Introduction to Robotics

Examination: Quiz III **Duration: 30min**

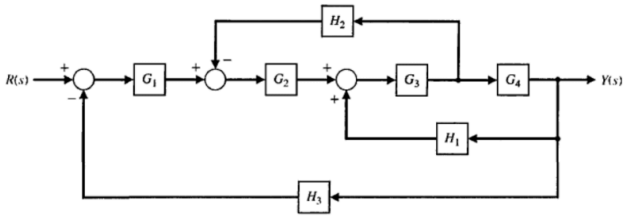
Semester : Summer 2026 **Full Marks: 15**

Name:	**ID:**	**Section:**

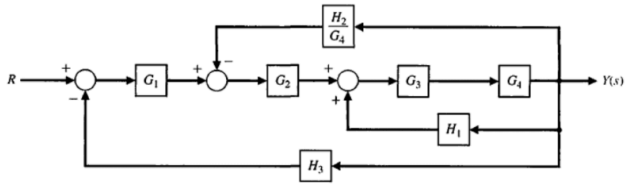
1	 A. **Reduce** the block diagram.	**10**
2	For a PI control system, the proportional and integral gain has been found to be 2.25 and 0.6 respectively. **Find the resulting gain and oscillation period for the given system. [2]**	**2**
3	 A system reaches its marginally stable state using only proportional control and generates the given graph. Using Ziegler-Nichols Rules, **find** the PID controller gains. **[3]**	**3**

Solution:

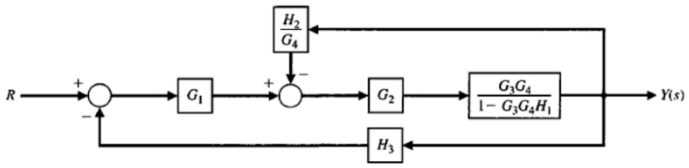
2. Block Diagram Solving



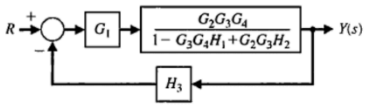
First, to eliminate the loop $G_3G_4H_1$, we move H_2 behind block G_4



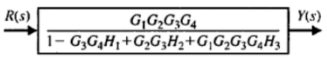
Eliminating the loop $G_3G_4H_1$ we obtain



Then, eliminating the inner loop containing H_2/G_4, we obtain



Finally, by reducing the loop containing H_3, we obtain



3. **Resulting gain, K = 5 (2.25/0.45) , Oscillation period = 2 (1.2/0.6)**

4. K = 0.5, P = 0.7, **Kp = 0.3, Ki = 2.85, Kd = 0.0875**